

Supplementary Exam

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Electricity and Electronics EBN 111

4 July 2022

Test Information

Maximum marks:	46	Full marks:	45
Duration of test:	105 min	Open/Closed book:	Closed
Total number of pages (this page included)		22	

Lecturers: Prof X. Ye, Prof J.D. le Roux, Mr V. Dube**Assistant Lecturers:** N. Matic, K. Mphaga, J. Thiselton, D. Wepener, L. Zheve**ECSA GA assessment**

The following ECSA graduate attributes are assessed at entry level in this question paper: GA1 and GA2.

Important

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Write your answers on the OCR forms provided. The question number **01** in this question paper corresponds to Row 01 of the OCR form, etc.
4. Read the instructions on the OCR form carefully.
5. Answers must include units where applicable.
6. Final answers must be written as real numbers and not as fractions. Use at least 4 significant figures to write numbers, including complex numbers.

INSTRUCTIONS**Do STEP 0 in the first 5 minutes of the session. (-5 to 0 min)****Do STEP 1 in the next 90 minutes of the test. (0-90 min)****Do STEP 2 within the next 15 minutes. (90-105 min)**

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR forms.

STEP 1 (0 min to 90 min): Do your own calculations on the question paper. Write down the answers on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (90 min to 105 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2022EBN111E02**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Make use of full stops as decimal points.

Section 1**Question 1 [2]**

A constant current of 400 mA passes through a resistor. The resistor converts 50 J of electrical energy to heat in 20 s. What is the potential difference v_R across the resistor?

01 v_R [2]

$$\Delta W = 50 \text{ J}$$

$$\Delta T = 20 \text{ s}$$

$$P = \frac{\Delta W}{\Delta T} = \frac{50}{20} = 2.5 \text{ W}$$

$$\text{but } P = v_e \times i_e$$

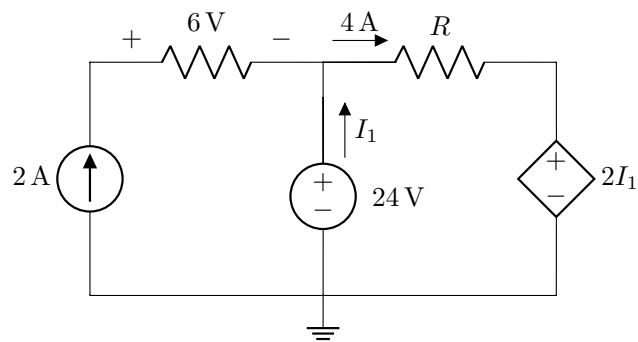
$$2.5 = v_e \times 400 \text{ mA}$$

$$v_e = \frac{2.5}{0.4} = \underline{\underline{6.25 \text{ V}}}$$

Question 2 [2]

Calculate R for the circuit shown below.

(The circuit contains sufficient information to solve the problem.)



02 R [2]

$$\text{KCL:} \quad 2 + I_1 = 4$$

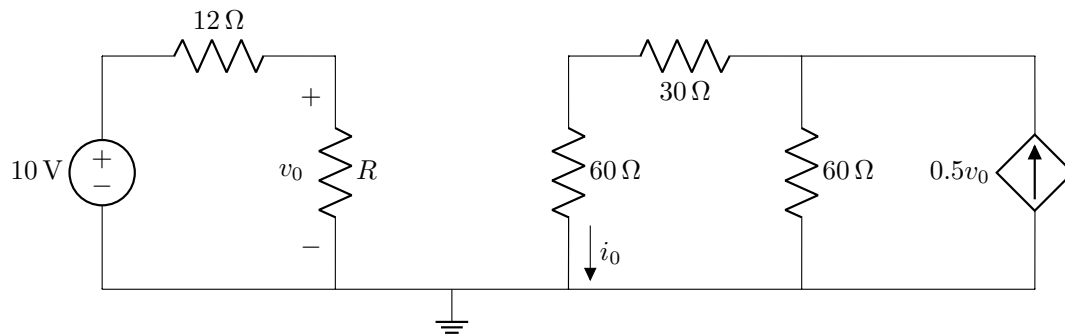
$$I_1 = 2 \text{ A}$$

$$\text{KVL:} \quad -24 + V_R + 2I_1 = 0$$

$$V_R = 24 - 2(2) = 20 \text{ V}$$

$$\therefore R = \frac{20}{2} = \underline{10 \Omega}$$

Questions 3 to 4 [2]

Determine R and i_0 if $v_0 = 8\text{ V}$.03 R [1]04 i_0 [1]

③ V-division

$$8 = \frac{10 \times R}{12 + R}$$

$$96 + 8R = 10R$$

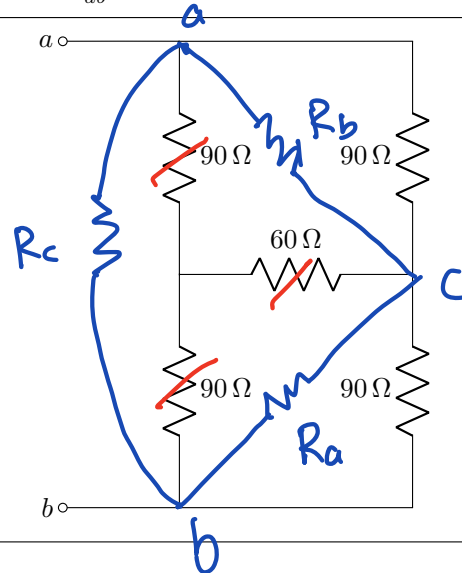
$$R = \frac{96}{2} = \underline{48\ \Omega}$$

④ I-division

$$i_0 = \frac{60 \times (0.5 \times 8)}{60 + 60 + 30}$$

$$= \underline{1.6\text{ A}}$$

Question 5 [2]

Determine the equivalent resistance R_{ab} between the two terminals $a - b$.05 R_{ab} [2]

$$R_a = 60 + 90 + \frac{60 \times 90}{90} = 210 \Omega.$$

$$R_b = 90 + 60 + \frac{60 \times 90}{90} = 210 \Omega.$$

$$R_c = 90 + 90 + \frac{90 \times 90}{60} = 315 \Omega.$$

$$= 315 \parallel (210 \parallel 90 + 210 \parallel 90)$$

$$= 90 \Omega.$$

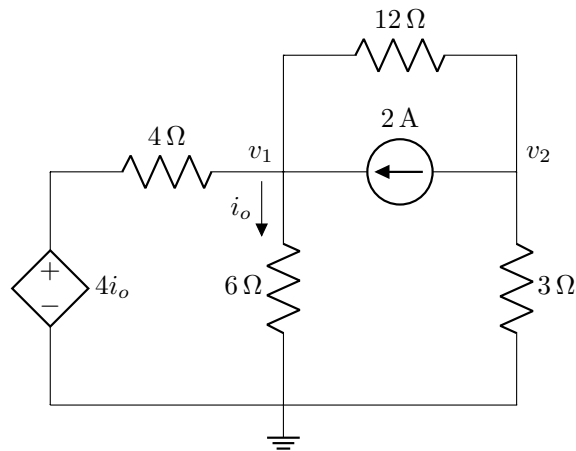
Questions 6 to 9 [4]

Use nodal analysis to set up two equations in the form:

$$av_1 - 1v_2 = b, \quad (\text{Node 1})$$

$$cv_1 + dv_2 = -24. \quad (\text{Node 2})$$

Calculate the unknown coefficients a , b , c and d .



06 a [1]

07 b [1]

08 c [1]

09 d [1]

$$i_o = \frac{v_1}{6}$$

$$\text{Node 1: } 2 = \frac{v_1}{6} + \frac{v_1 - v_2}{12} + \frac{v_1 - 4\left(\frac{v_1}{6}\right)}{4}$$

$$(\times 12) \quad 24 = 2v_1 + v_1 - v_2 + 3v_1 - 2v_1$$

$$24 = 4v_1 - v_2$$

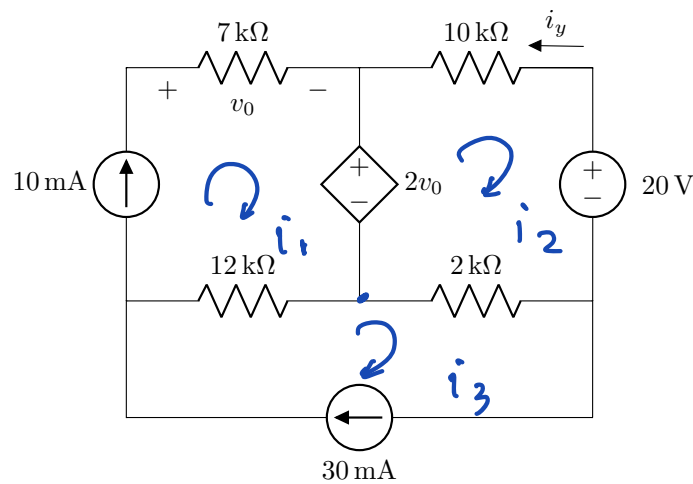
Node 2:

$$+2 + \frac{v_2}{3} + \frac{v_2 - v_1}{12} = 0$$

$$(\times 12) \quad 4v_2 + v_2 - v_1 = -24$$

$$5v_2 - v_1 = -24$$

Question 10 [2]

Use mesh analysis to determine i_y .10 i_y [2]

$$i_1 = 10 \text{ mA}$$

$$i_2 = -i_y$$

$$i_2 = 30 \text{ mA}$$

$$V_0 = 7 \text{ k}\Omega \cdot 10 \text{ mA} = 70 \text{ V}$$

$$\text{op2: } - \quad \circ - \quad k \cdot i_2 + 20 + 2k \cdot (i_2 - i_3) = 0$$

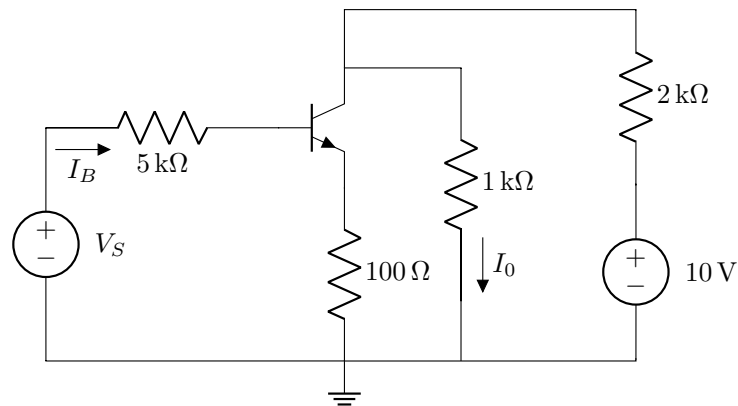
$$\Rightarrow 12k i_2 = -20 + 2V_0 + 2k i_3 = 180$$

$$\Rightarrow i_2 = 15 \text{ mA}$$

$$\Rightarrow i_y = -15 \text{ mA}$$

Question 11 [2]

Assume $V_{BE} = 0.7\text{ V}$ and $\beta = 100$. If $I_B = 25\text{ }\mu\text{A}$, determine I_0 .



11 I_0 [2]

$$\star I_C = 100 \times 25\text{ }\mu\text{A} = 2.5\text{ mA}$$

$$\star \frac{V_0 - 10}{2k} + \frac{V_0}{1k} + 2.5\text{ mA} = 0$$

$$(\times 2k) \quad V_0 - 10 + 2V_0 + 5 = 0$$

$$3V_0 = 5$$

$$V_0 = \frac{5}{3}\text{ V}$$

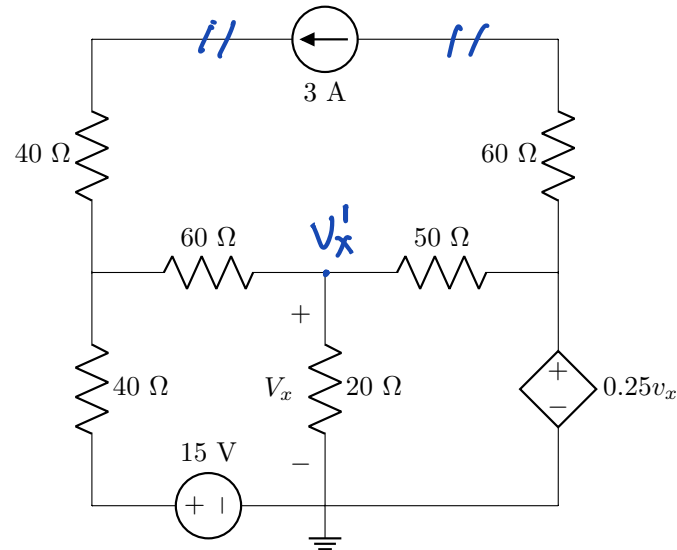
$$\therefore I_0 = \frac{\frac{5}{3}}{1k} = \underline{1.667\text{ mA}} \rightarrow$$

Total Section 1: [16]

Section 2

Question 12 [2]

Use superposition to determine V_x^1 , the exclusive contribution from the 15 V independent voltage source to V_x .



12 V_x^1 [2]

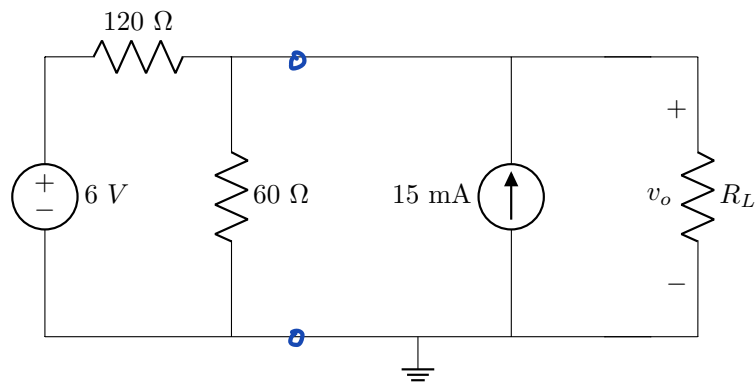
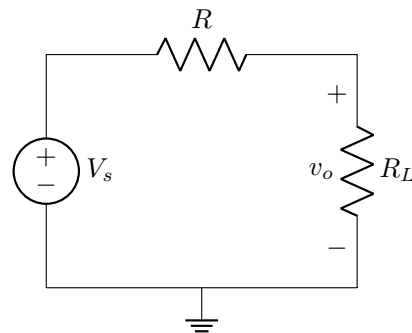
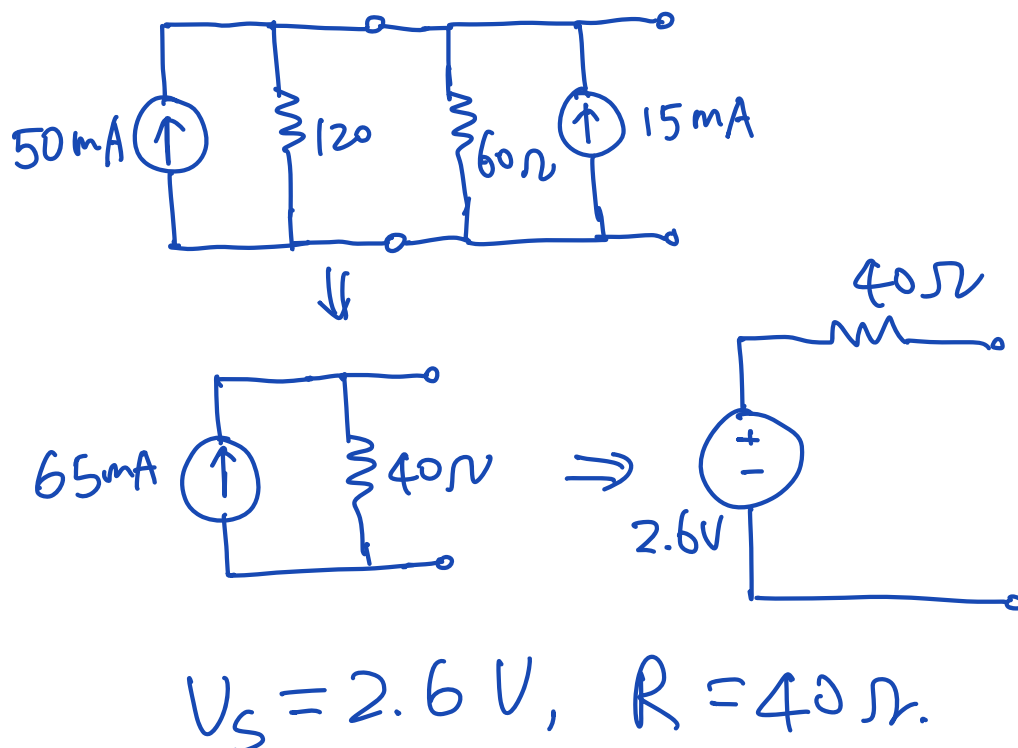
$$\frac{V_x - 15}{100} + \frac{V_x}{20} + \frac{V_x - 0.25V_x}{50} = 0, \quad \times 100$$

$$V_x - 15 + 5V_x + 2V_x - 0.5V_x = 0$$

$$\Rightarrow 7.5V_x = 15$$

$$V_x = 2V.$$

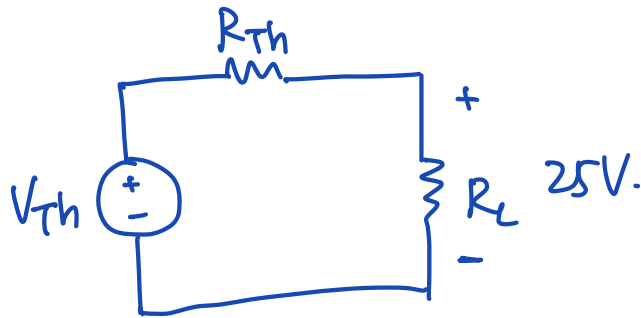
$$V_x = V_x^1 \text{ when } 3A \text{ is off}$$

Questions 13 to 14 [2]Use source transformation to transform Circuit A to Circuit B. Determine V_s and R .**Circuit A:****Circuit B:****13** V_s [1]**14** R [1]

Question 15 [2]

A Thévenin equivalent circuit is connected to a load resistor R_L , and the maximum power P_{max} is delivered to the load. If the voltage across R_L is 25 V and $P_{max} = 800$ mW, determine R_L .

15 R_L [2]



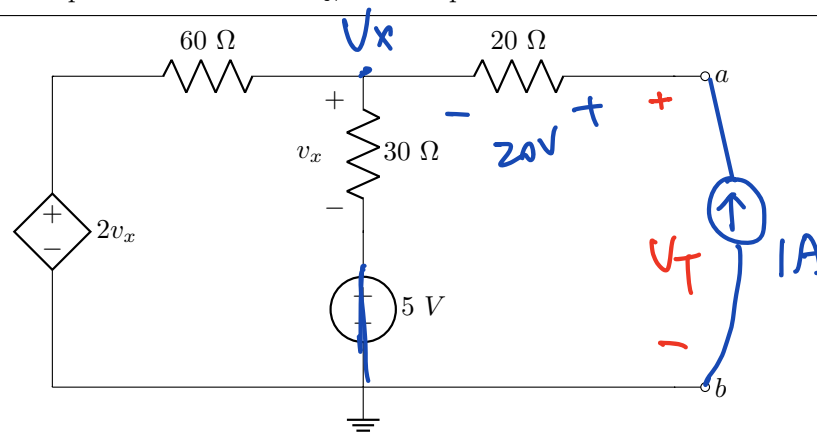
$$V_{Th} = 50 \text{ V.}$$

$$R_{Th} = R_L.$$

$$\frac{V_{Th}^2}{4R_L} = 0.8$$

$$\Rightarrow 4R_L = 3125$$

$$R_L = 781.25 \Omega.$$

Question 16 [2]Determine the Norton equivalent resistance R_N with respect to terminals $a - b$.**16** R_N [2]

$$\frac{V_x - 2V_x}{60} + \frac{V_x}{30} = 1$$

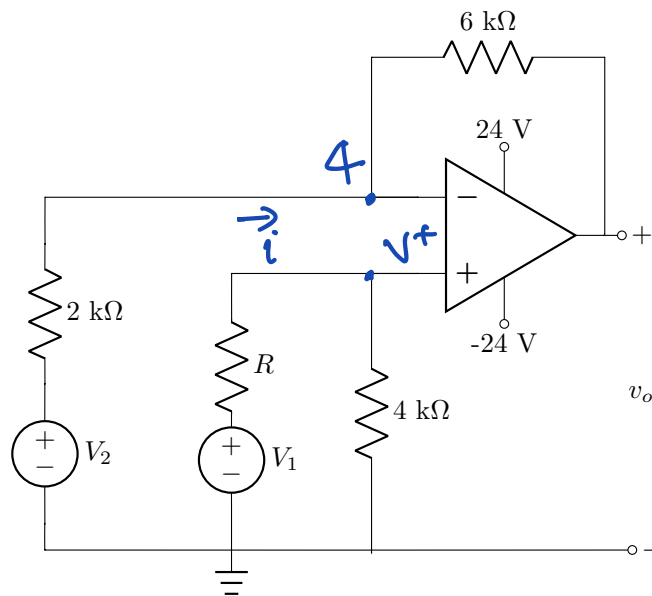
$$V_x - 2V_x + 2V_x = 60 \Rightarrow V_x = 60V$$

$$V_T = 20 \times 1 + V_x = 80V$$

$$\Rightarrow R_N = \frac{V_T}{1A} = 80\Omega.$$

Questions 17 to 18 [4]

Assume that the operational amplifier is ideal.



17 Determine v_o if $R=8\text{ k}\Omega$, $V_1=12\text{ V}$, and $V_2=2\text{ V}$. [2]

18 If $V_1=6\text{ V}$, $R=\infty\text{ k}\Omega$, find the minimum value of V_2 such that the operational amplifier is just about to enter the negative saturation. [2]

$$17. \quad V^+ = V_1 \cdot \frac{4}{8+4} = 4\text{ V.}$$

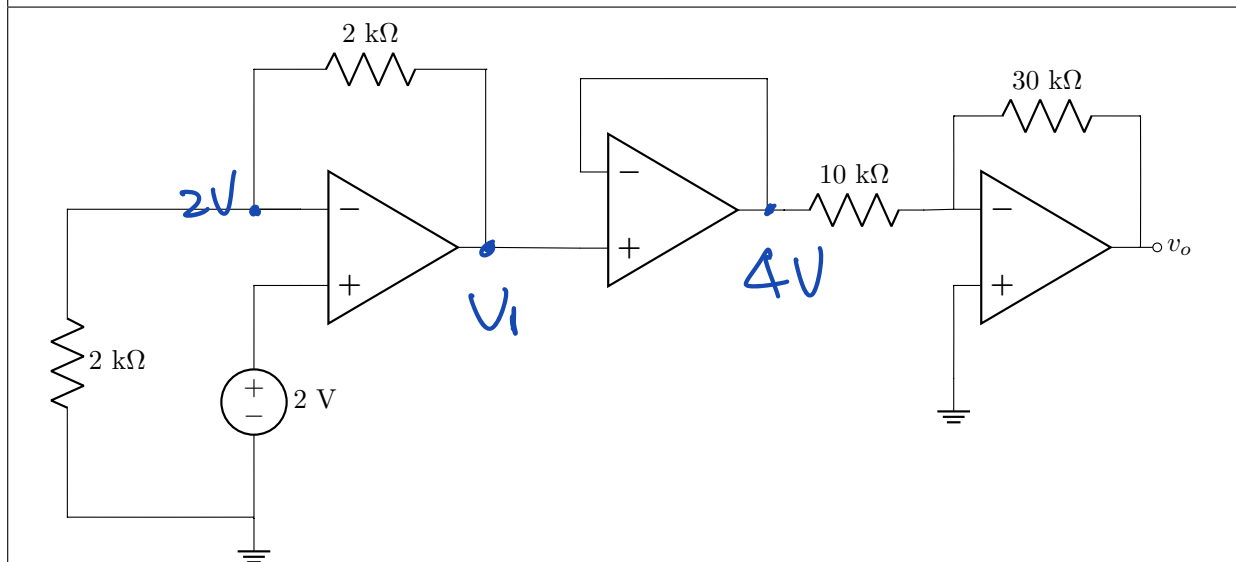
$$i = \frac{V_2 - 4}{2\text{ k}} = \frac{4 - V_o}{6\text{ k}}$$

$$\Rightarrow 3V_2 - 12 = 4 - V_o \Rightarrow \underline{V_o = 4 + 12 - 3V_2 = 10\text{ V.}} \rightarrow$$

18 $R \rightarrow \infty$, V^+ is grounded, Op Amp is inverting.

$$V_o = -3V_2 \leq -24$$

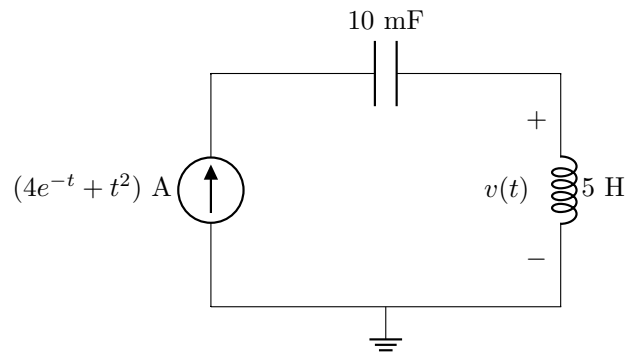
$$\Rightarrow V_2 \geq 8\text{ V.} \quad \underline{V_2^{\min} = 8\text{ V.}} \rightarrow$$

Question 19 [2]Assume that the operation amplifiers are ideal, determine v_o .**19** v_o [2]

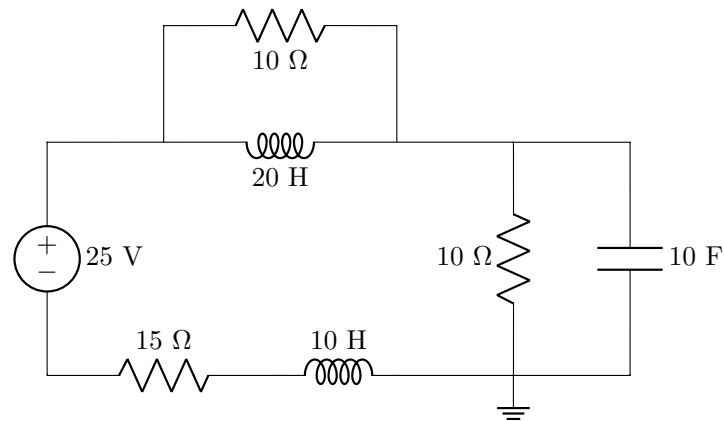
$$V_1 = 2 \left(1 + \frac{2k}{2k} \right) = 4V.$$

$$V_o = 4 \cdot \left(-\frac{30k}{10k} \right) = -12V.$$

Total Section 2: [14]

Section 3**Question 20 [2]**Determine $v(t)$ at $t = 2$ ms.**20** | $v(2 \text{ ms})$ [2]

$$\begin{aligned} v(t) &= L \frac{di}{dt} = 5 [4e^{-t} + t^2]' \\ &= 5 [-4e^{-t} + 2t] \\ v(2\text{ms}) &= 5 [-4e^{-2 \times 10^{-3}} + 2 \times 10^{-3} \times 2] \\ &= -19.94 \text{ V.} \end{aligned}$$

Question 21 [2]Under dc conditions, find the energy stored in the capacitor w_C .**21** w_C [2]

$$V_C = 25 \cdot \frac{10}{10+15} = 10 \text{ V}$$

$$\Rightarrow w_C = \frac{1}{2} C V^2 = \frac{1}{2} \times 10 \times 10^2 = 500 \text{ J}.$$

Question 22 [2]

Consider two sinusoid waveforms $v_1(t)$ and $v_2(t)$, where

$$v_1(t) = 10 \cos(3t - 60^\circ), \text{ and}$$

$$v_2(t) = -4 \sin(3t + 90^\circ).$$

The waveforms $v_1(t)$ and $v_2(t)$ can be represented by phasors $\mathbf{V}_1$ and $\mathbf{V}_2$, respectively.

22 Find $\mathbf{V}_T = \mathbf{V}_1 - \mathbf{V}_2$. [2]

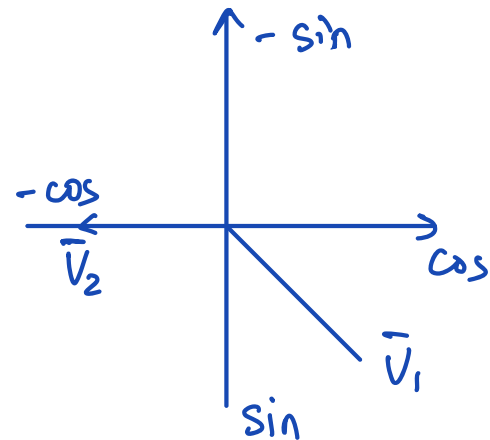
$$\phi_d = 120^\circ.$$

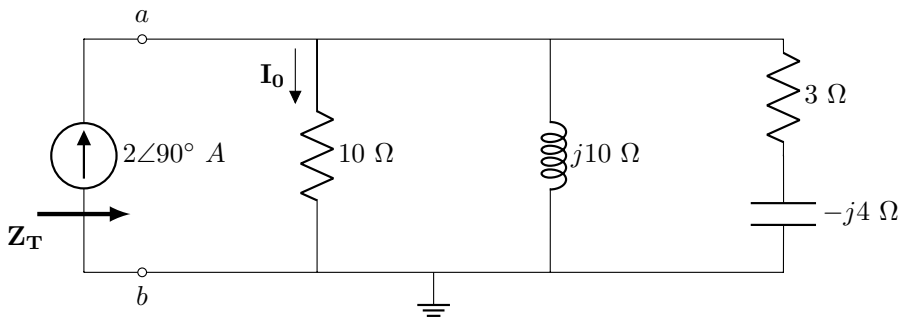
$$V_1 = 10 \angle -60^\circ$$

$$V_2 = 4 \angle -180^\circ$$

$$V_1 - V_2 = 9 - j5\sqrt{3}$$

$$= 12.49 \angle -43.8979^\circ$$



Questions 23 to 24 [4]	
Determine the impedance $\mathbf{Z_T}$ with respect to terminals $a - b$, and the current $\mathbf{I_0}$.	
	
23	$\mathbf{Z_T}$ [2]
24	$\mathbf{I_0}$ [2]

23

$$Z_T = 10 || j10 || (3 - j4)$$

$$= \frac{10 \times j10}{10 + j10} || (3 - j4) = (5 + j5) || (3 - j4)$$

$$= \frac{(5 + j5)(3 - j4)}{8 + j} = 4.385 \angle -15.255^\circ \Omega$$

$$= 4.231 - j1.154 \Omega$$

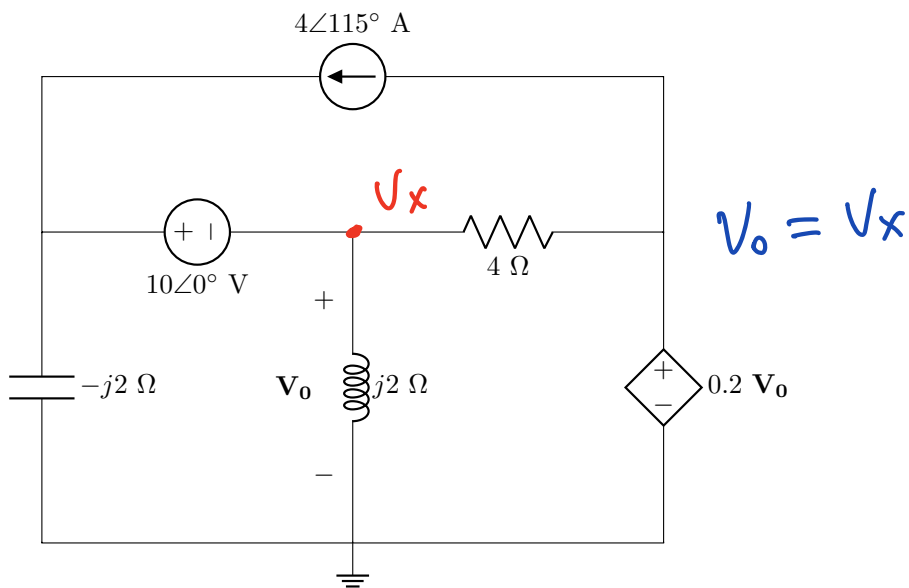
24

$$Z_1 = j10 || (3 - j4) = \frac{20}{3} - j\frac{10}{3}$$

$$I_0 = j2 \cdot \frac{Z_1}{Z_1 + 10} = j2 \cdot \frac{\frac{20}{3} - j\frac{10}{3}}{\frac{20}{3} - j\frac{10}{3} + 10}$$

$$= 0.2308 + j0.8462 \text{ A}$$

$$= 0.8771 \angle 74.7449^\circ \text{ A}$$

Question 25 [2]Find V_0^1 , the exclusive contribution from the independent voltage source to V_0 .25 | V_0^1 [2]

$$\frac{V_x + 10}{-j2} + \frac{V_x}{j2} + \frac{V_x - 0.2V_0}{4} = 0$$

$$\frac{V_x + 10}{-j2} + \frac{V_x}{j2} + \frac{0.8V_x}{4} = 0 \quad \xrightarrow{\times j4} \triangleright$$

$$-2(V_x + 10) + 2V_x + j0.8V_x = 0$$

$$\Rightarrow j0.8V_x = 20$$

$$\Rightarrow V_x = -j25 \text{ V.}$$

Questions 26 to 27 [4]	
Determine the Thévenin equivalent voltage V_{Th} , and the Thévenin equivalent impedance Z_{Th} with respect to terminals $a - b$.	
26	V_{Th} [2]
27	Z_{Th} [2]

26
$$V_{Th} = 2V_x \cdot \frac{-j10}{10-j10}$$

$$V_x = j8 \cdot (j2 \parallel 3 + j5) = j8 \cdot \frac{j2(3+j5)}{3+j7}$$

$$= -12.1379 + j1.6552 \text{ V}$$

$$\Rightarrow V_{Th} = 2V_x \cdot \frac{-j10}{10-j10} = -10.4828 + j13.7931 \text{ V.}$$

27 $V_x = 0$, add $V_T = 1\angle 0^\circ \text{ V}$, find I_T . $V_x = 0$

$$I_T = \frac{1}{10} + \frac{1}{-j10} = 0.1 + j0.1$$

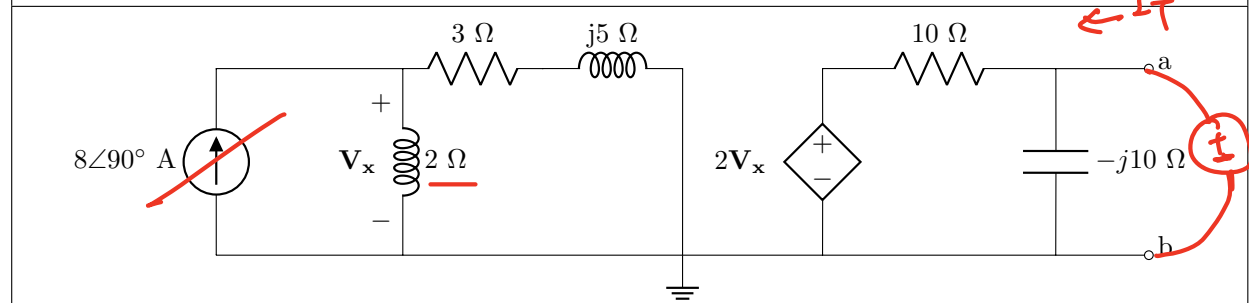
$$\Rightarrow Z_{Th} = \frac{1}{I_T} = 5 - j5 \Omega.$$

Total Section 3: [16]

Grand Total : [46]

Questions 26 to 27 [4]

Determine the Thévenin equivalent voltage V_{Th} , and the Thévenin equivalent impedance Z_{Th} with respect to terminals $a - b$.



26 V_{Th} [2]

27 Z_{Th} [2]

26 $V_{Th} = 2V_x \cdot \frac{-j10}{10-j10}$ (when 2Ω is read as a resistor)

$$V_x = j8 \cdot (2 \parallel 3+j5) = j8 \cdot \frac{2(3+j5)}{5+j5}$$

$$= -3.2 + j12.8 \text{ V.}$$

$$\Rightarrow V_{Th} = 2 \cdot (-3.2 + j12.8) \cdot \frac{-j10}{10-j10} = 9.6 + j16 \text{ V}$$

27 $V_x = 0$, add $V_T = 1\angle 0^\circ \text{ V}$, find I_T . $V_x = 0$.

$$I_T = \frac{1}{10} + \frac{1}{-j10} = 0.1 + j0.1$$

$$\Rightarrow Z_{Th} = \frac{1}{I_T} = 5 - j5 \Omega.$$

Total Section 3: [16]

Grand Total : [46]

PRACTICE ANSWER SHEET

Assessment ID:	2	0	2	2	E	B	N	1	1	1	E	0	2
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Enter complex numbers on OCR forms:										
5	∠	-	9	0	°				V	
6	.	9	∠	1	5	.	9	°	μ	A
1	2	-	<i>j</i>	8						
<i>j</i>	1	5							Ω	
-	1	.	5	+	<i>j</i>	8	.	2	m	V

Complex numbers can be entered in either **rectangular** or **polar** form.

Rectangular: The *j* is at the start of the imaginary number.

The *j* is in the CENTRE of the cell.

The *j* does NOT cross or touch cell boundaries!

Polar: The degree sign ° is directly after the angle value.

Round irrational numbers to a minimum of 4 significant figures.

Answer		Unit
SECTION 1		
01	$v_R =$	
02	$R =$	
03	$R =$	
04	$i_0 =$	
05	$R_{ab} =$	
06	$a =$	No unit required
07	$b =$	No unit required
08	$c =$	No unit required
09	$d =$	No unit required
10	$i_y =$	
11	$I_0 =$	

Round irrational numbers to a minimum of 4 significant figures.

Answer		Unit
SECTION 2		
12	$V_x^1 =$	
13	$V_s =$	
14	$R =$	
15	$R_L =$	
16	$R_N =$	
17	$v_o =$	
18	$V_2^{min} =$	
19	$v_o =$	
SECTION 3		
20	$v(2 \text{ ms}) =$	
21	$w_C =$	
22	$\mathbf{V_T} =$	
23	$\mathbf{Z_T} =$	
24	$\mathbf{I_0} =$	
25	$\mathbf{V_0^1} =$	
26	$\mathbf{V_{Th}} =$	
27	$\mathbf{Z_{Th}} =$	